THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $(-5, -2)$	Reflection in the y-axis changes the sign of the abscissa.
2	**(d)** $3x + 4$	Linear means degree exactly 1.
3	**(c)** 3	It is the larger of the two powers.
4	**(b)** $4x^2 + 12xy + 9y^2$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(b)** 48	Chord $= 2\sqrt{r^2 - d^2}$.
6	**(c)** 84	Use Heron's formula.
7	**(b)** $\frac{11}{50}$	Relative frequency of the outcome.
8	**(b)** 53	$2 \times 25 + 3$.
9	**(c)** If a number is divisible by 3, then it is divisible by 6.	Interchange the hypothesis and the conclusion.
10	**(d)** 44	$\frac{27 \times 57 + 39 \times 35}{66} = 44$.
11	**(a)** 2	Carries out of the columns: [1, 0, 1, 0].
12	**(a)** 144°	One part $= 360^\circ \div 15 = 24^\circ$.
13	**(d)** No	$1(-3) + (2)(2) = 1$, not equal to 2.
14	**(a)** 242 cm^2	$2\pi rh$ with $r = 3.5$, $h = 11$.
15	**(b)** 5	Apply the distance formula.
16	**(d)** -10	Put x in $p(x) = 0$.
17	**(b)** $\frac{3}{11}$	Multiply by 100 and subtract.
18	**(c)** $9x^2 - 12xy + 4y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
19	**(b)** 15	$d = \sqrt{r^2 - (\text{half-chord})^2}$.
20	**(b)** 21	$s = \frac{a+b+c}{2}$.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$PQ = \sqrt{(3)^2 + (4)^2}$ (1); $PQ = 5$ units (1).	1+1
22	$3x - 18 = 0$ gives $x = 6$ (1); The graph meets the x-axis at $(6, 0)$ (1).	1+1
23	$x = 0.\overline{3}$, $10x - x = 3$ (1); $x = \frac{1}{3}$ (1).	1+1
24	Use the square identity (1); $= 9x^2 + 30xy + 25y^2$ (1).	1+1
25	Half-chord $= \sqrt{26^2 - 10^2} = 24$ cm (1); Chord $= 48$ cm (1).	1+1
26	$s = 21$ cm (1); $A = \sqrt{s(s-a)(s-b)(s-c)} = 84$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 64$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
28	$x = -1$: $y = 1$; $x = 0$: $y = 3$ (1); $x = 1$: $y = 5$; $x = 2$: $y = 7$ (1); Slope $= 2$, y-intercept $= 3$ (1).	1+1+1
29	$x = 0.\overline{36}$, $100x = 36.\overline{36}$ (1); $99x = 36$ (1); $x = \frac{4}{11}$ (1).	1+1+1
30	Use the square identity (1); $= 4x^2 + 12xy + 9y^2$ (1); LHS $= (2+3)^2 = 25$, RHS $= 25$ (1).	1+1+1
31	$OA = OB$ so $\angle OAB = \angle OBA$ (1); $\angle OAB = \frac{180^\circ - 70^\circ}{2} = 55^\circ$ (1); Angle on the major arc $= \frac{70^\circ}{2} = 35^\circ$ (1).	1+1+1
	OR: $s = 21$ cm (1); $A = 84$ cm^2 (1); $h = \frac{2A}{15} = \frac{168}{15} = \frac{56}{5}$ cm (1).	1+1+1
32	Total $= 12$ (1); $P(\text{red}) = \frac{1}{3}$ (1); $P(\text{not red}) = \frac{2}{3}$; $P(\text{black or white}) = \frac{2}{3}$ (1).	1+1+1
	OR: $7 + (n-1)(6) = 151$ (1); $(n-1) \times (6) = 144$ (1); $n = 25$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (3, 2)$ (1); Midpoint of $BD = (3, 2)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
34	$C(x) = 12x + 60$ (1); $C(5) = 120$ (1); $12x + 60 = 168$ (1); $x = 9$ (1); Hence the number of notebooks is 9 (1).	1+1+1+1+1
35	Remainders repeat after 6 steps (1); $\frac{1}{7} = 0.\overline{142857}$ (1); Block length $= 6$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
36	Use $(a - b)^2 = a^2 - 2ab + b^2$ (1); $a = 2x$, $b = 3y$ (1); Squares: $4x^2 + 9y^2$ (1); Cross term: $-12xy$ (1); $= 4x^2 - 12xy + 9y^2$; at $x = 3$, $y = 1$ the value is 9 (1).	1+1+1+1+1
	OR: Distance of first chord from centre $= \sqrt{13^2 - 5^2} = 12$ cm (1); Distance of second chord $= \sqrt{13^2 - 12^2} = 5$ cm (1); Same side: subtract the distances (1); Distance between chords $= \lvert 12 - 5 \rvert = 7$ cm (2).	1+1+1+2
37	Outer area $= \frac{22}{7} \times 21^2 = 1386$ m^2 (1); Inner area $= 616$ m^2 (1); Track area $= 770$ m^2 (1); Cost $= 770 \times 5$ (1); $=$ ₹3850 (1).	1+1+1+1+1
	OR: Total trials $= 100$ (1); Prime: $2, 3, 5$: $\frac{53}{100} = \frac{53}{100}$ (1); Greater than 4: $\frac{30}{100} = \frac{3}{10}$ (1); Less than 3: $\frac{37}{100} = \frac{37}{100}$ (1); All six frequencies add to 100, so relative frequencies add to 1 (1).	1+1+1+1+1
38	Fares ₹240, ₹280, ₹320 (1); Common difference ₹40, so an AP (1); $t_n = a + (n - 1)d$ with $a = 240$, $d = 40$ (1); $t_{12} = 240 + 11 \times 40$ (1); Fare ₹680 (1).	1+1+1+1+1
	OR: Let the integers be $n, n + 1, \ldots, n + 2$ (1); Sum $= 3n + 3 = 3(n + 1)$ (1); A multiple of 3, proved (1); Check: $8 + 9 + 10 = 27 = 3 \times 9$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{7}{25}$	Relative frequency.
2	**(a)** 17th term	$3n+2=53$ gives $n=17$.
3	**(a)** False	Counterexample: 9 is divisible by 3 but not by 6.
4	**(a)** 33	$\frac{14\times31+61}{15}=33$.
5	**(b)** 6	Divisors of 44: 1, 2, 4, 11, 22, 44.
6	**(d)** 7	Opposite angles are equal.
7	**(b)** 4	Put x=0: $y=4$.
8	**(d)** 671 cm^2	$2\pi r(r+h)$ with $r=3.5$, $h=27$.
9	**(b)** $(2,2)$	Use $\left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$.
10	**(a)** -1	Substitute $x=-k$.
11	**(d)** $\sqrt{5}$	$2^2<5<9$
12	**(b)** $x^2+4x-21$	$(x+a)(x+b)=x^2+(a+b)x+ab$
13	**(c)** 37°	Half of the central angle.
14	**(d)** 616	Find r from $C=2\pi r$, then use $A=\pi r^2$.
15	**(c)** $\frac{5}{9}$	Count vowels among all the letters (repeats counted).
16	**(b)** 19	AP with $a=4$, $d=3$: $t_6=4+5\times3$.
17	**(c)** $a=-3,\ b=2$	$a^2=9>b^2=4$.
18	**(a)** 69	$\frac{70\times6+65\times6+75\times3}{15}=69$.
19	**(a)** 2	Pairs: $(90,82)\to(82,8)\to(74,8)\to(66,8)\to(58,8)\to(50,8)\to(42,8)\to(34,8)\to(26,8)\to(18,8)\to(10,8)\to(8,2)\to(6,2)\to(4,2)\to(2,2)\to(2,0)$.
20	**(c)** 20°	Adjacent angles add up to 180°.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Even outcomes: $2,4,6$ with frequencies $22,16,16$ (1); $P=\frac{54}{100}=\frac{27}{50}$ (1).	1+1
22	$t_8=25$ (1); $3n+1=37$ gives $n=12$, so yes (1).	1+1
23	Converse: If a number is divisible by 3, then it is divisible by 6. (1); it is false, counterexample: 9 is divisible by 3 but not by 6 (1).	1+1
24	Total of first group $=27\times67=1809$; Total of second group $=45\times35=1575$ (1); Combined average $=\frac{3384}{72}=47$ (1).	1+1
25	Replace (m,n) by $(n,m-n)$ re-ordered so the larger is first; $(54,30)\to(30,24)\to(24,6)\to(18,6)\to(12,6)\to(6,6)\to(6,0)$ (1); 6 subtractions, gcd $=6$ (1).	1+1
26	$\angle C=\angle A=88^\circ$ (opposite angles); $\angle B=180^\circ-88^\circ=92^\circ$ (adjacent angles) (1); $\angle D=\angle B=92^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Converse: If a number is divisible by 3, then it is divisible by 6. (1); Counterexample: 9 is divisible by 3 but not by 6 (1); It satisfies the hypothesis of the converse but not its conclusion, so the converse is false (1).	1+1+1
28	Pure part in the first $=110\times\frac{38}{100}=41.8$ mL; Pure part in the second $=96.8$ mL (1); Total $=138.6$ mL in 330 mL (1); Concentration $=42\%$ (1).	1+1+1
29	Replace (m,n) by $(n,m-n)$ re-ordered so the larger is first (1); $(92,72)\to(72,20)\to(52,20)\to(32,20)\to(20,12)\to(12,8)\to(8,4)\to(4,4)\to(4,0)$ (1); 8 subtractions, gcd $=4$ (1).	1+1+1

30	Opposite angles are equal; $2x + 8 = x + 16$ gives $x = 8$ (1); Angles $A = C = 24^\circ$ (1); $B = D = 180^\circ - 24^\circ = 156^\circ$ (1).	1+1+1
31	Slope of the first two points $= \dfrac{-3-(1)}{-1-(-3)} = -2$ (1); Slope between $(-1,-3)$ and $(1,k)$ must also be -2: $\dfrac{k-(-3)}{2} = -2$ (1); $k = -3 - 2 \times 2 = -7$ (1).	1+1+1
	OR: $l = \sqrt{r^2+h^2} = 50$ cm (1); CSA $= \pi rl = 2200$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 9856$ cm^3 (1).	1+1+1
32	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x-(-1))^2 + 16 = (x-5)^2 + 16$ (1); $x = 2$, point $(2, 0)$ (1).	1+1+1
	OR: Substitute: $11 = 6 + k$ (1); $k = 5$ (1); Line: $y = 2x + 5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	Totals: 1320, 1222, 1672 (1); Grand total $= 4214$ for 86 students (2); Average $= 49$ (2).	1+2+2
34	Rule: new $x = 2x - 2$ (1); Values: 5, 8, 14, 26, 50, 98 (2); Final $x = 98$ (2).	1+2+2
35	$AD = 14$ cm (D is the midpoint) (1); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 19$ cm (1); Hence $AE = EC$ (2).	1+1+1+2
36	Rewrite as $y = -2x + 3$; table: $x = -1 \Rightarrow y = 5$, $x = 0 \Rightarrow y = 3$, $x = 1 \Rightarrow y = 1$, $x = 2 \Rightarrow y = -1$ (1); Plot the four points; they lie on one straight line (1); Slope $= -2$ (1); y-intercept $= 3$, i.e. the point $(0, 3)$ (2).	1+1+1+2
	OR: $2\pi r = 44$, so $r = \dfrac{44 \times 7}{2 \times 22} = 7$ cm (1); CSA = area of the sheet $= 44 \times 25 = 1100$ cm^2 (2); Volume $= \pi r^2 h = \frac{22}{7} \times 7^2 \times 25 = 3850$ cm^3 (2).	1+2+2
37	$AB = \sqrt{225} = 15$ (1); $BC = \sqrt{144} = 12$ (1); $CA = \sqrt{81} = 9$ (1); Perimeter $= 36$ units (2).	1+1+1+2
	OR: $y(0) = -12$, $y(1) = -8$, $y(2) = -4$ (1); Crosses x-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= 4$ (1); Slope is positive: the graph rises (1).	1+1+1+1+1
38	First $= 5^2 - 3 = 22$ (rational) (1); Second $= 25 + 3 + 10\sqrt{3}$ (1); $= 28 + 10\sqrt{3}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1
	OR: Use $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1); $a = x$, $b = 2$ (1); $3a^2b = 6x^2$, $3ab^2 = 12x$ (1); $= x^3 + 6x^2 + 12x + 8$ (1); Terms checked (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** -9	$\frac{-2-(7)}{5-(4)} = -9$.
2	**(b)** 2200 cm^2	πrl with the given values.
3	**(c)** $(6, 0)$	Ordinate is 0 on the x-axis.
4	**(b)** 7	Put $x = 2$.
5	**(c)** irrational	Use the rules for sums and products of rational and irrational numbers.
6	**(d)** 10609	Write $103 = 100 + 3$
7	**(c)** 147°	Angles on the two arcs are supplementary: $180^\circ - t$.
8	**(d)** 176	Distance in one turn = circumference.
9	**(a)** $\frac{3}{8}$	Use the 8 equally likely outcomes.
10	**(b)** 47	$a + (n-1)d = 5 + 14 \times (3)$.
11	**(d)** 9	List the factors of 36 in pairs; the count is 9 (odd, so 36 is a perfect square).
12	**(b)** 27%	$\frac{60 \times 15 + 120 \times 33}{180} = 27\%$.
13	**(c)** 2	$146 = 3 \times 48 + 2$.
14	**(b)** 28 cm	Diagonals bisect each other, so $AC = 2 \times OA$.
15	**(c)** 1	Rewrite as $y = x + \ldots$; slope $= -\frac{a}{b}$.
16	**(c)** 1617 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.
17	**(a)** Quadrant II	Use the sign rules for the quadrants.
18	**(d)** $(4, 0)$	Put $y = 0$.
19	**(d)** $\frac{1}{6}$	Use the repeating-decimal method.
20	**(b)** $(x + 4)$	$a^2 + 2ab + b^2 = (a + b)^2$

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$y = 0$: $4x = 36$ gives $x = 9$; point $(9, 0)$ (1); $x = 0$: $6y = 36$ gives $y = 6$; point $(0, 6)$ (1).	1+1
22	$l = \sqrt{r^2 + h^2} = 25$ cm; CSA $= \pi rl = 550$ cm^2 (1); Volume $= \frac{1}{3}\pi r^2 h = 1232$ cm^3 (1).	1+1
23	$\left(\frac{-3+(9)}{2}, \frac{7+(1)}{2}\right)$ (1); Midpoint $= (3, 4)$ (1).	1+1
24	$p(2) = 11$ (1); $p(-2) = -5$ (1).	1+1
25	Divide 1 by 6 (1); $= 0.1\overline{6}$ (1).	1+1
26	$104 = 100 + 4$ (1); $104^2 = 10816$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	$x = 0.1\overline{6}$; $10x = 1.\overline{6}$ and $100x = 16.\overline{6}$ (1); $90x = 15$ (1); $x = \frac{1}{6}$ (1).	1+1+1
28	Need two numbers with sum 8 and product 12 (1); They are 2 and 6 (1); $= (x + 2)(x + 6)$ (1).	1+1+1
29	$\angle A + \angle C = 180^\circ$; $5x + 5 = 180$ (1); $x = 35$ (1); $\angle A = 80^\circ$ (1).	1+1+1
30	$s = 13$ cm (1); $A = \sqrt{(s-a)(s-b)(s-c)(s-d)}$ (1); $A = 36$ cm^2 (1).	1+1+1
31	Walk $= 200 - 80 - 50 = 70$ (1); $P(\text{walk}) = \frac{7}{20}$ (1); $P(\text{not bus}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1).	1+1+1
	OR: $a = 6$, $d = 5$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{15} = 615$ (1).	1+1+1
32	A false statement needs only one counterexample (1); Take 2 (1); 2 is prime and even, so the statement is false (1).	1+1+1
	OR: Sum of 10 observations $= 10 \times 53 = 530$ (1); New sum $= 530 + 64 = 594$ (1); New mean $= \frac{594}{11} = 54$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	$\angle C = 180^\circ - 75^\circ = 105^\circ$ (1); $\angle D = 180^\circ - 88^\circ = 92^\circ$ (1); $\angle DCE = \angle A = 75^\circ$ (1); $\angle DAF = 180^\circ - 75^\circ = 105^\circ$ (1); Reasons: opposite angles supplementary; exterior angle property (1).	1+1+1+1+1
34	Outer length $= 34$ m, outer breadth $= 24$ m (1); Outer area $= 816$ m^2 (1); Lawn area $= 600$ m^2 (1); Path area $= 216$ m^2 (1); Units stated (1).	1+1+1+1+1
35	Number of letters $= 11$ (1); $P(\text{vowel}) = \frac{4}{11}$ (1); $P(S) = \frac{4}{11}$ (1); $P(A \text{ to } M) = \frac{5}{11}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1
36	Monthly savings form an AP (1); $t_{12} = 300 + 11 \times 50 = 850$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{12} = \frac{12}{2}[600 + 550]$ (1); Total ₹6900 (1).	1+1+1+1+1
	OR: Let the integer be $2m + 1$; its square $= 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1$ is odd (1); Converse: if n^2 is odd then n is odd (1); For $7^2 = 49$ (odd), n is odd (1); If n were even, n^2 would be even, so the converse is true (2).	1+1+1+2
37	January food $= 1539$ (1); February food $= 1960$ (1); February is higher by 421 (1); The larger total outweighs the smaller share in February (2).	1+1+1+2
	OR: Digits picked (from the right): 6, 9, 6, 5 (1); Sum $= 6 + 9 + 6 + 5$ (2); Final $s = 26$ (the digit sum) (2).	1+2+2
38	In $\triangle ABC$, $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$; in $\triangle ADC$, $SR \parallel AC$ and $SR = \frac{1}{2}AC$ (1); So $PQ \parallel SR$ and $PQ = SR$: a pair of opposite sides equal and parallel, so PQRS is a parallelogram (1); $PQ = SR = 8.5$ cm and $QR = PS = 7.5$ cm (1); Perimeter $= 17 + 15 = 32$ cm (2).	1+1+1+2
	OR: $F = \frac{9}{5} \times 50 + 32 = 122$ (1); For $F = 212$: $\frac{9}{5}C = 180$, so $C = 100$ (2); Slope $\frac{9}{5}$: each 1°C rise raises F by 1.8° (2).	1+2+2

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